

User Manual and Technical Documentation
*Laboratory-Scale , Three-Phase Grid-Forming Inverter for
Islanded Microgrids*

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Table of Contents

1	Introduction.....	3
2	Safety precautions.....	4
3	Hardware overview.....	5
4	Technical Specifications	6
5	Connections	6
6	Dashboard	7
7	Operating procedure	10
8	Troubleshooting.....	11

List of Figures

Figure 1:Font panel and side cooling panel	5
Figure 2: Side cooling panel	5
Figure 3:Back panel with USB port.....	5
Figure 4: Font panel.....	5
Figure 5: Connection setup of the device	6
Figure 6: Dashboard	7

List of Tables

Table 1: Technical specifications	6
Table 2: Troubleshooting in system.....	11

1 Introduction

Purpose

This prototype was developed as a laboratory-scale Grid-Forming Inverter (GFM) to demonstrate stable three-phase AC power generation for islanded microgrids. It serves as an open-architecture platform for research, education, and the development of advanced inverter control strategies.

What is a Grid-Forming Inverter?

A Grid-Forming Inverter is a voltage-source inverter capable of independently establishing and regulating the voltage and frequency of an electrical system. Unlike conventional grid-following inverters, it does not require an existing grid reference and can operate autonomously in islanded microgrids.

Why This Setup Was Developed

This system was developed to address the lack of accessible Grid-Forming Inverter research platforms in Sri Lanka. It provides a practical hardware testbed for validating control algorithms, demonstrating inverter operation, and supporting future research on renewable energy integration and islanded microgrids.

Intended Users

This prototype is intended for undergraduate and postgraduate students, researchers, educators, and engineers working in the fields of power electronics, renewable energy systems, and microgrid control. It can be used for laboratory demonstrations, controller development, and experimental validation of Grid-Forming Inverter technologies.

2 Safety Precautions

- Wear insulated rubber shoes and appropriate personal protective equipment (PPE) when operating the inverter.
- Disconnect all power sources before making or modifying any electrical connections.
- Verify the polarity and voltage level of the DC input supply before energizing the system.
- Ensure all electrical connections are secure and properly insulated before operation.
- Never short-circuit the inverter output terminals or the DC input terminals.
- Do not touch exposed conductors or circuit components while the inverter is energized.
- Use insulated probes and properly rated measuring instruments for voltage and current measurements.
- Ensure the inverter is properly grounded before operation.
- Do not exceed the rated 240 V DC input voltage or 5 A maximum input current.
- Operate the inverter only within its specified load rating.
- Keep metallic objects and conductive materials away from the energized circuit.
- Allow sufficient time for the DC-link capacitors to discharge before handling the power circuit after shutdown.
- Ensure adequate ventilation around the inverter to prevent overheating of power electronic components.
- Do not operate the system if any component appears damaged, loose, or overheated.
- Verify the correct operation of the gate driver and protection circuits before applying the DC supply.
- Ensure the emergency power disconnect is easily accessible during operation.
- Keep the work area dry and free from moisture to reduce the risk of electric shock.
- Do not leave the inverter operating unattended during testing.
- Operate the system only under the supervision of qualified personnel or a laboratory instructor.
- In the event of abnormal noise, excessive heating, smoke, or unusual odor, immediately disconnect the DC supply and inspect the system before restarting.

3 Hardware Overview



Figure 1: Front panel and side cooling panel



Figure 2: Side cooling panel



Figure 4: Front panel



Figure 3: Back panel with USB port

4 Technical Specifications

Table 1: Technical specifications

Parameter	Value
Input DC voltage (Maximum)	240 V
Input current (DC maximum)	5 A
Output AC voltage (Phase to neutral RMS)	84.5 V
Frequency	50 Hz
Switching frequency	10 kHz

5 Connections

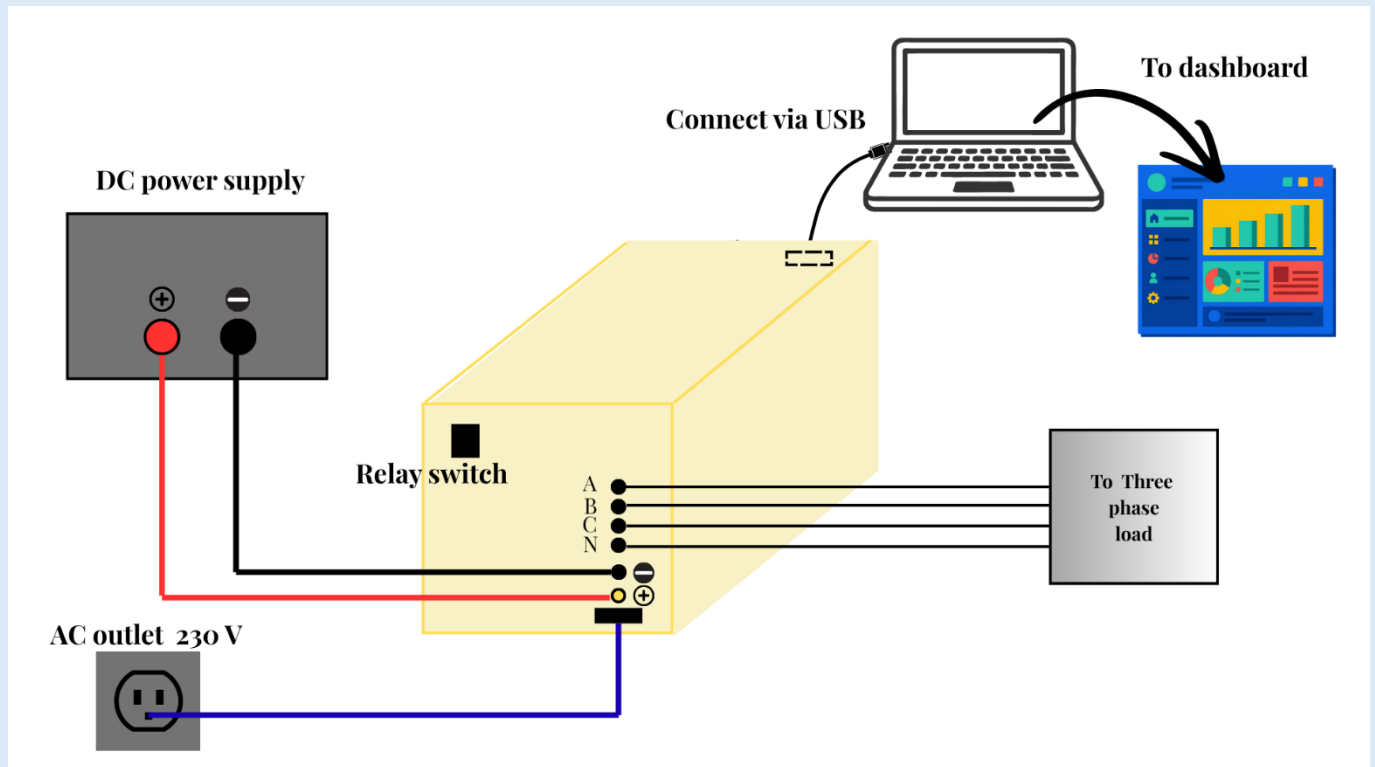


Figure 5: Connection setup of the device

6 Dashboard

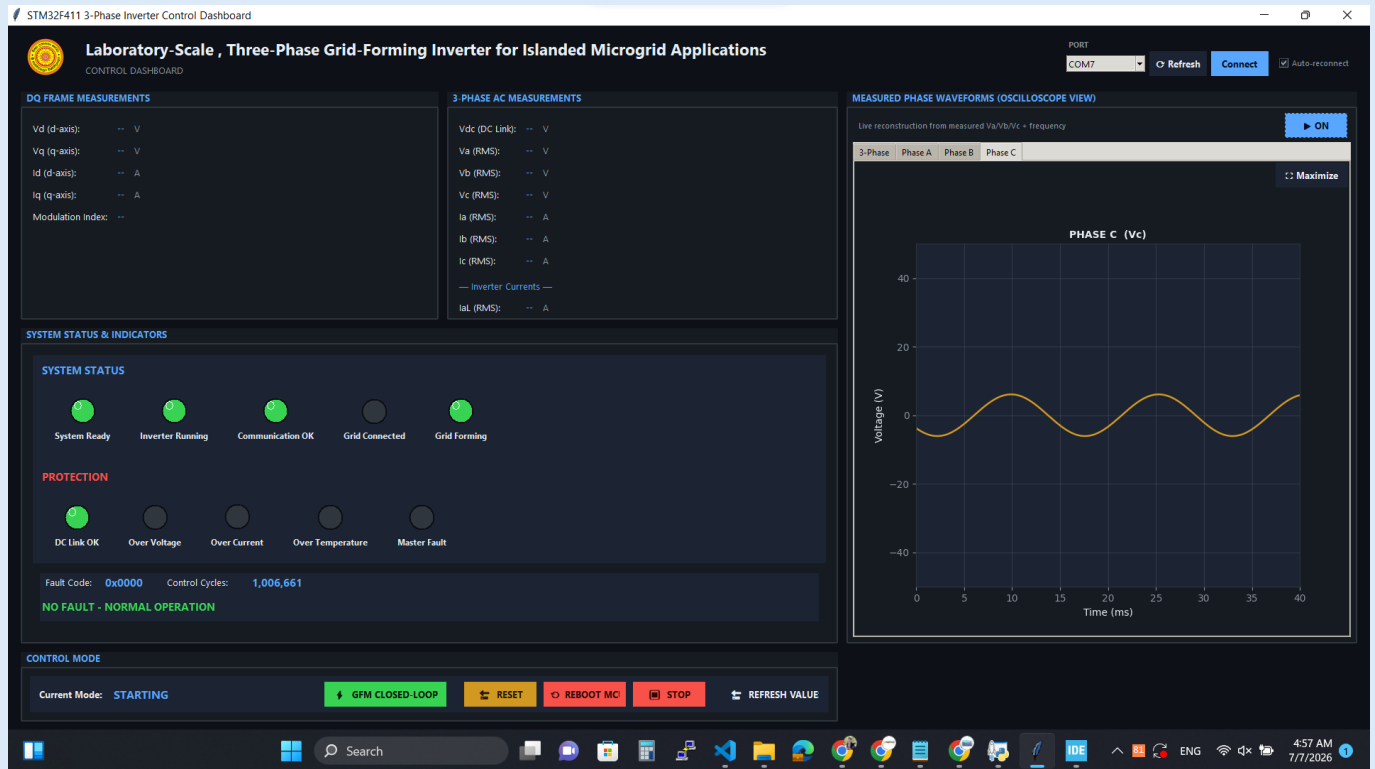


Figure 6: Dashboard

1. DQ Frame Measurements

This section displays the electrical quantities in the synchronous rotating reference frame.

- Vd (d-axis Voltage): Direct-axis voltage component.
- Vq (q-axis Voltage): Quadrature-axis voltage component.
- Id (d-axis Current): Direct-axis current component.
- Iq (q-axis Current): Quadrature-axis current component.
- Modulation Index: PWM modulation index used by the inverter.

These parameters are useful for evaluating the performance of the control algorithm and verifying the operation of the dq control loops which is useful in research purposes.

2. Three-Phase AC Measurements

The dashboard continuously displays the measured AC and DC electrical parameters.

Measured values include:

- DC Link Voltage (Vdc)
- Phase A RMS Voltage (Va)
- Phase B RMS Voltage (Vb)

- Phase C RMS Voltage (V_c)
- Phase A RMS Current (I_a)
- Phase B RMS Current (I_b)
- Phase C RMS Current (I_c)
- Inverter Output Current

These measurements are updated in real time from the microcontroller and provide instant feedback regarding inverter operating conditions.

3. Real-Time Waveform Display

The right panel displays reconstructed voltage waveforms of the inverter output.

Features include:

- Three-phase waveform visualization
- Individual Phase A, Phase B, and Phase C viewing
- Oscilloscope style plotting
- Zoom (Maximize) function
- Live waveform update
- Waveform enable/disable option

The graphical display enables users to verify waveform quality, phase relationships, frequency stability, and sinusoidal output characteristics during operation.

4. System Status Indicators

The dashboard provides visual LED indicators representing the operating status of the inverter.

Available status indicators include:

- System Ready
- Inverter Running
- Communication OK
- Grid Connected
- Grid Forming

Green indicators represent normal operating conditions, while gray indicators denote inactive states.

These indicators allow the operator to quickly determine the current operating mode of the inverter.

5. Protection Monitoring

The protection section continuously monitors critical fault conditions.

Monitored protection functions include:

- DC Link Status
- Over Voltage
- Over Current
- Master Fault

Whenever a protection event occurs, the corresponding indicator changes state, allowing immediate identification of the fault source.

The dashboard displays:

- Fault Code
- Controller Cycle Count
- Current Fault Message

During normal operation, the dashboard displays:

NO FAULT – NORMAL OPERATION

6. Communication Interface

The upper right section allows the user to establish communication with the inverter controller.

Functions include:

- COM Port Selection
- Refresh Available Ports
- Connect to STM32
- Auto-Reconnect Option

Once connected, the dashboard begins receiving measurement data continuously through serial communication.

7. Control Panel

The bottom section provides operational controls for the inverter.

Available controls include:

- GFM Closed-Loop – Starts the inverter in Grid-Forming closed-loop mode.
- Reset – Clears fault conditions and resets controller variables.

- Reboot MCU – Restarts the STM32 microcontroller.
- Stop – Stops inverter operation safely.
- Refresh Values – Requests updated measurement values from the controller.

7 Operating Procedure

Step 1 : Connect the DC supply

Step 2 : Connect the Auxiliary supply to power outlet

Step 2 : Connect testing loads to phases according to

Step 3 : Turn on DC power supply

Step 4 : Once DC power reach the expected limit , turn on relay switch

Step 5 : Observe dashboard.

Step 6 : Measure timeline data from CSV file.

When shutting down ;

Step 1 : Turn off relay switch

Step 2 : Turn off DC supply

Step 3: Disconnect system from power outlets (DC power supply and auxiliary supply)

Step 4: Remove load connections

8 Troubleshooting

Table 2: Troubleshooting in system

Problem	Solution
No output voltage at the inverter output	Verify the 240 V DC input supply, ensure the relays are energized, and check that the controller is powered.
Output frequency is not 50 Hz	Verify the controller firmware configuration and confirm that the reference frequency is set to 50 Hz.
MOSFETs overheating	Ensure adequate heat sinking, verify dead-time settings, and check for excessive load current or short circuits.
Not hearing 'Tik' sound when switching on When turning on the auxiliary switch.	Check relay connections
Relay does not switch ON	Verify the relay supply voltage, controller output signal, and relay driver circuit connections.
Dashboard is not displaying data	Verify the USB connection, COM port selection, baud rate, and dashboard software configuration.
Inverter shuts down unexpectedly	Check for overcurrent, loose connections, overheating, or insufficient DC input voltage.